Muhammad Usama

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Introduction

Research Assistant with **2 years** in multimodal generation, Parametric Shape Modeling, and 3D Reconstruction, supported by **3+ years** of Machine learning Engineering/software engineering experience. Published in leading international venues and experienced in building scalable ML systems for real-world use.

Publications

** equally contributing first author*

- **NURBGen**: **Muhammad Usama***, Mohammad Sadil Khan*, Didier Stricker, and Muhammad Zeshan Afzal (2026). *NURBGen:High-Fidelity Text-to-CAD Generation through LLM-Driven NURBS Modeling.* **AAAI 2026** **[Paper]**
  - First LLM-driven **NURBS-based** Text-to-CAD framework.
  - Introduced *PartABC*, a **300K** multi-modal CAD dataset with NURBS and captions.
  - Proposed hybrid analytic–NURBS representation for robust CAD generation.
  - Achieved **state-of-the-art** fidelity on CAD generation benchmarks.
- **MARVEL-40M+**: Sankalp Sinha*, Mohammad Sadil Khan*, **Muhammad Usama**, Shino Sam , Didier Stricker , Sk Aziz Ali, and Muhammad Zeshan Afzal. (2025). *MARVEL-40M+:Multi-Level Visual Elaboration for High-Fidelity Text-to-3D Content Creation.* **CVPR 2025** **[Paper]**
  - Introduced the largest 3D captioning dataset (**40M+** annotations, **8.9M** assets).
  - Integrated domain-specific **metadata** to reduce VLM **hallucinations**.
  - Developed MARVEL-FX3D, a two-stage text-to-3D generation pipeline.
- **DreamCAD**: Mohammad Sadil Khan, **Muhammad Usama**, Rolandos Alexandros Potamias, Didier Stricker, Muhammad Zeshan Afzal, Jiankang Deng, Ismail Elezi (2026). *DreamCAD:Scaling Multi-modal CAD Generation using Differentiable Parametric Surfaces.* **CVPR 2026** **[Submitted]**
  - Developed a multi-modal CAD generation framework using differentiable parametric surfaces **(Bézier Surface)**.
  - Introduced **CADCap-1M**, the largest text-to-CAD captioning dataset.
  - Achieved **state-of-the-art** results on text-, image-, and point-to-CAD benchmarks.

Experience

- **Student Research Assistant** — Kaiserslautern, Rheinland-Pfalz, Germany
  German Research Center (DFKI) — *Nov 2023 – Present*
  - **Automated captioning pipeline** : Built an end-to-end pipeline to auto-caption **8.9M 3D assets** using vision-language models (VLMs); orchestrated multi-view renders and metadata extraction with Blender.
  - **Fine-tuning LLMs** : Researched and implemented **fine-tuning** strategies for large language models (LLMs), including Qwen-3B and Qwen-7B, using parameter-efficient methods such as **LoRA**.
  - **Long Context Fine-tuning** : Explored techniques for long-context fine-tuning of LLMs, evaluating approaches like **LongLoRA** and other context-extension strategies to enhance model performance on long-sequence tasks.
  - **Ethical filtering** : Designed an LLM-based ethical filtering layer with prompt engineering to detect, redact, and rewrite problematic captions; reduced ethical concerns by **80%** relative to the unfiltered output.
  - **Pose-guided image generation**: Gained in-depth knowledge of diffusion models through extensive research, evaluated existing methodologies and continuously analyzed ongoing advancements in pose-guided image generation.
  - **Visual Machine Learning**: Built backend **microservices** for a visual ML platform to create, train, and deploy models, supporting both the DFKI cluster and on-prem MindGarage CPU nodes.

- **Software Engineer — Machine Learning Infrastructure (Worked Remote from Karachi, Sindh, Pakistan)** Redmond, WA, United States (Remote)
  Retrocausal *Feb 2023 – Oct 2023*
  - **Software Packaging**: Led the software packaging for client deployment, tailoring systems to diverse factory environments. This played a key role in expanding our client base and securing **venture capital funding in late 2023**, marking a significant career milestone that reflects the global scale and recognition of my work.
  - **System Scalability**: Improved system scalability and efficiency by replacing continuous polling with **Redis-** and **SQS**-based messaging, enabling on-demand data loading and significantly reducing API calls for long-running and continuous tasks.
  - **Labelling Tool**: Built an efficient object labelling method for the system, enabling accurate labelling of 2D static and 3D objects in depth and RGB frames. Designed an intuitive UI, resulting in time savings and enhanced product appeal.
  - **On-Premises Architecture Design and Development**: At a major client's request, migrated the architecture from cloud to **on-premises** with minimal performance loss, replacing AWS with **Localstack** and alternatives to Firebase and other cloud services, and built a packaging script for seamless installation on client PCs.
  - **Thin Web Client** : Developed a lightweight web client version to replace the bulky desktop application UI. Implemented **WebRTC** for seamless streaming and data channel communication.
- **Software Engineer** Karachi, Sindh, Pakistan
  Aga Khan Development Network Digital Health Resource Centre (AKDN) *Jul 2020 – Mar 2022*
  - **Full-Stack Development**: Built a scalable healthcare system (records, scheduling, billing, inventory, payments) for the largest hospital network in Karachi and Nairobi using **.NET**, **NestJS**, and **Angular**.
  - **Deployment Automation & Containerization**: Modernized deployment workflows by introducing **Docker**, **Docker Compose**, and **GitHub Actions**, reducing manual steps and production errors.

Education

- **Technische Universität Kaiserslautern-Landau (RPTU)** Kaiserslautern, Rheinland-Pfalz, Germany
  Master of Science in Computer Science *Oct 2023 – Present*
  - **Master's Thesis** (in progress): Developing a zero-shot multimodal 3D classification and retrieval framework
- **National University of Computer and Emerging Sciences (NUCES)** Karachi, Sindh, Pakistan
  Bachelor of Science in Computer Science *Aug 2018 – Jul 2022*

Skills

- **Programming Languages**: Python, C/C++, Java, JavaScript, C#, Dart
- **ML Architecture**: VLM-LLM Inference, Transformers, Diffusion Model, Stable Diffusion, LORA
- **Software Engineering Frameworks** : .NET Core, Django, Spring Boot, ExpressJs, NestJs, Flutter, Java Android Development
- **ML Libraries** : PyTorch, PyTorch Lightning, Python OCC, vLLM, Blender Python,
- **Interpersonal** : Creative Thinker, Problem solver, Collaborative Engineer and Researcher, Quick and Adaptive Learner

Languages

English (C1), Deutsch (A2), Urdu (Native)

Hobbies

- **Sports**: Playing table tennis and cricket.
- **Travelling and learning**: Visited multiple cities across Greece, Saudi Arabia, Germany, France, Luxembourg, Singapore and Pakistan exploring local culture and history.